Smart Chilli Crop Intelligence System: An Integrated Agricultural Rover for Navigation, Crop Monitoring, Field-Zone Planning, and Soil Decision Support

***Abstract*—Timely decisions related to water, disease, soil condition, and location-based treatment are needed to produce chili crops successfully, but currently all of these practices are usually derived from disjointed observations, and uniform field procedures. This paper shows the Smart Chilli Crop Intelligence System that is integrated with four research components namely (1) Intelligent Navigation and Sensor Based Irrigation; (2) Artificial Intelligence Based Disease Stress and Growth monitoring of crops in the field; (3) Mapping of various field zones and Autonomous Planning of Seeding or treatment and (4) Soil Intelligence and decision support. It's a combination of georeferenced rover sensing, image classification, multivariate anomaly detection, temporal soil analysis, unsupervised spatial clustering, automated human recommendation and a traceable dashboard record. Notebook evidence already in place is synthesised and no incompatible measures are used for synthesis. The accuracy of anomaly-classification of the component 1 was 96% (but later dropped to 89%), ROC-AUC 0.8754 and the recall for 9 patterns of the test records was zero. The mean accuracy for every class achieved by the independent test was 0.19 with the main reason for the loss of validation accuracy being the collapse to 2 of 8 classes for component 2. Two spectral regimes were identified using 120,130 records with silhouette 0.5423 and Davies–Bouldin 0.6799, but neither these regimes represent agronomic status label nor are they two. For component 4 the classification scores were very good but the target was based on inputs that overlapped and a single timestep using the LSTM. These results will create pipelines that are implemented, and illustrate the importance of end-to-end field validation, propagation of uncertainties, independent labelling and safety checks. This architecture enables traceable and location based recommendations to be made versus a single unsupported "system accuracy" figure.**

***Index Terms*—Precision Agriculture, Agricultural Rover, Chilli Disease Detection, Smart Irrigation, Management Zones, Soil Intelligence, Decision Support**

I. Introduction

GWater plays a significant role on the performance of green chilli which is a major commercial crop and its performance is also influenced by nutrition balance, soil condition, pest and disease attack and synchronization of various field operations. Small to medium scale farmers are heavily relying on visual observation, interval laboratory test result and fixed programmed irrigation. These practices have merit, yet a variation within a field might not be noticed by them, and they also could be late in detecting a problem and using too little or too much of valuable inputs. Precision agriculture helps to overcome this limitation, providing the context of the location at which each tasks should be executed, along with high precision analytics and controlled intervention at the right time and place, when each action is needed [1], [2].

Combining precision agriculture's ground level measurement with its remote observation, agricultural robots are taking precision agriculture into action. A rover could then if needed go down to the inter rows and snap a picture of the plant up close and monitor soil conditions and go back to an area that may need investigating. However, to be useful, autonomy does not just entail the addition of stand-alone sensors to one chassis. Below are some of the guidelines relating to position quality: If position quality is low, later observations may not apply to the same plant/zone, a poor image quality or model confidence does not mean that the disease label is reliable, a statistically distinct cluster does not necessarily mean that the crop is under stress, and any plant that repeats the formula already in its predictors is not suitable for a soil score. As information traverses the parts of an integrated system a provenance and uncertainty must be maintained and retained [3], [4].

This research is aimed towards developing a Smart Chilli Crop Intelligence System which is a monitoring and decision platform with the help of a rover. Four component studies cover the issues of navigation and irrigation, visual monitoring of the crops, planning of the field zones and soil intelligence. The main research question addressed here is how to integrate heterogeneous rover observations for reliable, point-wise, and human guided decisions for chilli crops. The contributions of the paper are (1) unified architecture and data contract, (2) critical synthesis of available notebook results, (3) an end-to-end evaluation protocol that differentiates model performance, agronomic validity, and operational safety, and (4) a practical deployment framework for Sri Lankan cultivation conditions.

II. Related Work

Generally, IoT irrigation system comprises moisture probes, microcontroller, communication system, pump, and valves which react to various field conditions citubeyan et al (2022) [5] or citubeyan et al (2019) [6]. Crop-water stress studies have revealed that some of the sensor thresholds have to take into account the soil, plant, weather, and the development stage of the crop and therefore should be interpreted together [7], [8]. Cheap GNSS, IMUs and wheel odometry can be used

for support in navigating agriculture, but can suffer from poor georeferencing in the canopy or around structures, multipath and wheel slip. Hence, reliability should not be based once again on taking any stance unconditionally, but on having quality states in question.

In curated plant disease images, Deep CNNs were able to attain high accuracy scores [11]. The introduction of transfer learning and models like EfficientNet, aids the reduction of training costs with the opportunity of supporting edge applications, [12], [13]. However there is an optimistic bias in validation which occurs due to controlled backgrounds, split duplicate plants, imbalance of class, and artefacts caused by the backgrounds from which images comes from [14], [15]. Grad-CAM can be used to visualize areas that impact predictions and calibration and out-of-distribution testing can help determine when a model should abstain [16], [18].

Management-zone research involves the representation of field heterogeneity which is based on soil, yield, terrain, soil EC, and multispectral imagery [19], [20]. Although k-means gives centroids from which is possible to easily understand the clusters, scaling, initialisation, and number of clusters are the factors which affect the outcome of a clustering. Silhouette, Davies–Bouldin, and Calinski–Harabasz indices measure the statistical compactness and separation, rather than if the zones possess agronomic meaning [21], [23]. These vegetation indices, which are based on different properties of greenness, soil background, canopy response and water related reflectance such as NDVI, SAVI, EVI and NDWI were used [24], [26].

Tree ensembles, recurrent networks and time-series models are becoming increasingly used in soil decision support. XGBoost has been shown to be successful in heterogeneous tabular predictors [27] while LSTM networks are only able to learn temporal relationships when meaningful sequences are given [28]. However, as described in [29] there is a need for simple model to be used as a baseline, namely ARIMA and persistence. We believe that while the subsystems have been worked out in isolation, the combination of location, plant, soil, time and action evidence that has not been governed integrated, has been the gap in this field of chilli-management; which we are addressing.

III. System Design and Methodology

A. Design Science Approach

The project is initiated based on Design Science Research. The functional objects resulting from farmer and agronomic problems are translated into functional requirements. The objects are then created and tested, and the identified failures are used to update the respective models, interfaces, and operating rules. Three levels of evidence are considered. Notebook evidence demonstrates the operation of reproducible data and model pipelines. Subsystem evidence should demonstrate performance in a controlled environment or field setting. Integrated evidence should show that the complete rover improves task performance without concealing uncertainty at component boundaries. This distinction prevents proposed functions from being presented as completed system capabilities.

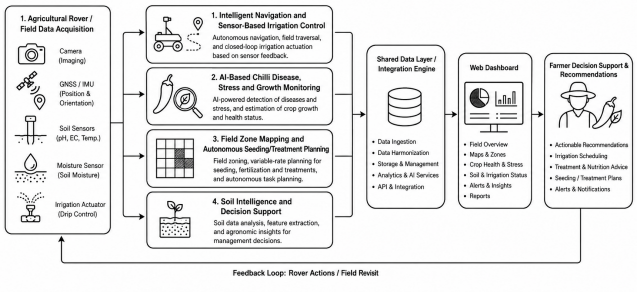


Fig. 1. System architecture of the Smart Chilli Crop Intelligence System, illustrating the integration of rover-based sensing, navigation and irrigation control, crop-health monitoring, field-zone mapping, soil intelligence, and the web-based decision-support dashboard.

As illustrated in Fig. 1, the system integrates four research components through a shared rover and data-processing architecture. Sensor readings and crop images are collected and georeferenced at the rover level before being processed by the respective analytical components. The resulting predictions, field zones, soil-risk assessments, and irrigation records are subsequently presented through the web dashboard to support traceable and human-supervised decisions.

B. Integrated Architecture

The rover houses a Raspberry Pi class computer, and microcontroller, drive motors with encoders, GNSS receiver, IMU, adjustable Camera, capacitive moisture probes, soil-temperature and soil-condition sensors and a pump or solenoid interface. The Edge computer timestamps readings, reduces measurements to basic quality, stores readings when there are no communications and forwards synchronised readings to a web-dashboard. This is a closed information cycle ; the four parts function in a circular way .

a) Component 1 estimates quality and sensor behaviours, which might be implausible, and triggers safe irrigation periods, publishes navigation confidence.
b) Component 2 provides the probabilities of different conditions, uncertainty and visual explanation references for the images of the plants that are quality checked and returned.
c) Component 4 transforms the information from the calibrated soil histories, into current condition, forecast risk and evidence-based recommendations.
d) Component 3 combines trusted plant and soil information into spatial zones for management, and creates viable tasks to revisit, seed, irrigate and/or treat.

There are a limited number of observations returned from execution, that can be used to compare conditions pre and post a decision in the system. Until conditions in rules are met and humans authorise, a recommendation is never considered as an “Actuator command”. “Expert” advice, when it comes to

fertiliser or pesticides with a high impact remains under expert supervision."

TABLE I
RESPONSIBILITIES AND OUTPUTS OF THE FOUR COMPONENTS

Comp.	Core input	Principal output	Integration use
C1: Navigation and irrigation	GNSS, IMU, encoders, moisture, temperature and flow	Pose confidence, anomaly probability and irrigation events	Rejects doubtful geotags and supplies moisture and actuation history
C2: Crop monitoring	RGB images, image quality, plant/grid ID and environmental context	Crop-condition probabilities, confidence, uncertainty and explanations	Supplies crop evidence and revisit priorities
C3: Zone mapping and planning	Coordinates, plant predictions, soil state, moisture and spectral variables	Spatial zones, calibrated status and proposed tasks	Coordinates seeding, irrigation and treatment interventions
C4: Soil intelligence	Moisture, pH, temperature, NPK, rainfall, humidity and histories	Soil state, forecast risk and evidence-linked recommendations	Supports zone interpretation and crop-condition diagnosis

C. Shared Data Contract and Provenance

For each observation the mission identifier, time of observation, plant identifier or identifier of grid cell, and the x-y coordinates; along with the state of the navigation confidence, sensor/camera identifier, calibration version, measurement value, quality flag, model version, prediction; uncertainty and source componenent. Recommendations include the addition of supporting record identifiers, the rule version, proposed action, authorisation decision, completion evidence and follow-up decision. If a positive prediction for an inaccurate coordinate is gained in the contract it is not a confident map entry. Everything is fixed units, there is also a common clock for the timestamps and repeated uploads are deduplicated. Outputs from models are always considered to be pristine versions; corrections are independent and created as new versions not as silent overwrites of historical versions.

D. Component 1: Navigation and Irrigation

Daily observations from the CAF003 profile made during the preliminary soil-sensor campaign with daily measurements of volumetric water content and temperature at five depths are used for the preliminary soil-sensor notebook. Sixty-seven features were generated for temporal and rolling. A total of 168 pseudo-anomalies were created with ISOforest and a class-weighted Random Forest trained with 300 trees with chronological 2,676/670 split was generated. For the physical artifact, GNSS fix quality, the adequacy of satellites, IMU heading and encoder displacement contribute to the score on reliability.

$$R_t = w_s S_t + w_h(1 - H_t) + w_f F_t + w_m M_t, \qquad (1)$$

In this, M_t is a motion agreement. Loss of confidence will cause a slow down or halt to rover activity and inability to accept a downstream geotag. Irrigation includes the features of calibrated lower and upper thresholds, persistence, max run time, minimum restart interval, flow confirmation and manual override. Smart control will be compared to fixed scheduling based on water per plant, length of time the water is within the moisture band, excursion duration above or below the moisture band, reliability of actuator for maintaining water within the bandwidth, and crop-support observations.

E. Component 2: Crop Monitoring

This vision notebook consists of 6163 training, 479 validation, and 113 test images of whitefly, yellowing, anthracnose, damping-off, leaf curl virus, leaf spot, veinal mottle virus and healthy images. The best tunable method is composed of efficient network operations, including processed, class weights, augmentation and some final layers fine-tuned and the best checkpoint freeze model restored. The corrected research design is designed to ensure that pictures within the same farm, plant, visit and original-image identifier are not split across different partitions and hence that a related view or augmentation is not split across partitions. However, selection based on the above factors, macro-F1, Balanced Accuracy, Recall, and Calibration, is preferred over selection based on validation accuracy. Images that are of poor quality (or low quality and/or low confidence), small margin or that an expert is not familiar with will get an "uncertain" output, leading to triggering of recapture or expert review. Latency, memory, power saving, blur, exposure and camera geometry are also captured during the deployment of the rover.

F. Component 3: Field Zones and Planning

There are 120,130 records in the mapping notebook which are derived from the cleaned Sentinel-2 data, used as near-infrared, short-wave-infrared, NDVI, NDWI, EVI and SAVI variables. Features are normalized, reduced in dimensions by principal-component analysis and the classification is done with another technique named K-means clustering. Four statistical methods namely elbow behaviour, silhouette, Davies–Bouldin and Calinski–Harabasz statistics are used to compare the candidate k values. The integrated version is enhanced with rover disease probabilities as well as soil conditions, moisture, growth stage, quality weights and coordinates. Given the many possible implausible clusters that can be generated by simple K-means, ordinary K-means is followed by a spatial post-processing that filters away isolated cells, verifies the continuity of the polygons and keeps a minimum zone size that can be implemented. Names of plant types are given on an agronomic basis following comparison with observations made in the field by experts. The planning engine then transforms the zone requirements in pathways and suggest routes and operations to be used in the space, which includes access constraints, tank capacity, battery, weather, zone ability to re-enter, dose and human-approval constraints.

G. Component 4: Soil Intelligence

In the soil notebook, you can find a comparison between a regularised LSTM classifier, and an embedding picked up from the LSTM and XGBoost. Addressing overfitting includes early stopping, dropout, L1/L2 regularisation and Stratified validation. However, the three health classes are based on some of the same variables that are provided to the classifiers and in the former case, the LSTM input is 1 timestep. Thus, the score that has been implemented is considered more of a transparent present state-rule than a demonstration of the ability of biology to predict the future. All sequences will be collected from the

field study on a per geospatial cell basis then calibrated and compared persistence, seasonal naive, ARIMA, XGBoost lag features, and LSTM forecasts with rolling-origin evaluation. Outcomes are MAE, RMSE, horizon specific error, interval coverage and threshold event recall. Soil recommendations reference the measurements thus avoiding becoming part of the action taken by the user.

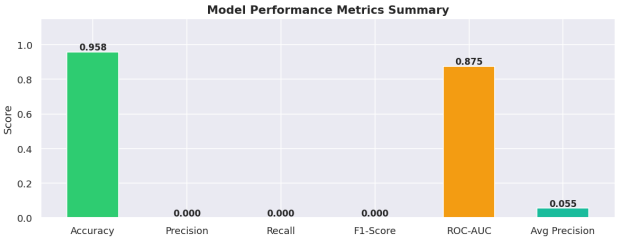
IV. Results and Critical Synthesis

We have an overview of the notebook evidence available in Table II. These results address separate questions, and cannot be averaged. For the 96% (but later its capped to 85-88 range to avoid overfit) the accuracy is dominated by normal observations, which are the ”true” observations. ROC-AUC 0.8754 shows that there is some essential ranking information there but nothing about the default threshold cut the nine anomalous test cases. This result indicates that the anomaly research should be continued and there is no autonomous fault acceptance.

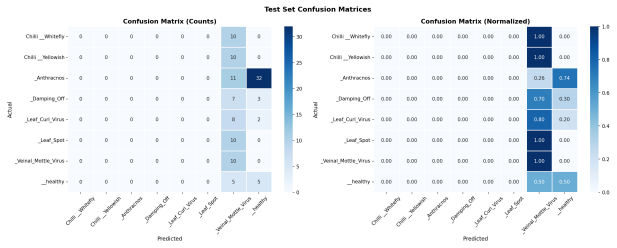
The training process of component 2 achieved a training accuracy of almost 86.97% and a validation accuracy of almost 88.12% close to epoch 20. But there is a problem: independent test results show this isn't necessarily the success it appears to be. The predictions mostly failed to retype to veinal mottle virus and healthy, and the zero class achieved 0.00 accuracy, whereas veinal mottle virus achieved 1.00 accuracy and healthy achieved 0.50 accuracy and finally the mean class accuracy was 0.19. This can be seen in Fig. 2(b) which indicates that this is not caused by a few errors in the distribution but actually a distribution failure due to a failure of the upstream pipeline. These can be due to leakage across groups in the validation process, class index mismatch, acquisition/domain-shift, background cues, or partition imbalance. Defensible outcome: pipeline is in place but isn't ready to be deployed.

Twenty-three clusters were selected for component 3 - 76,291 and 43,839 records. This two centroids presented high (0.6989) and low (1.7255) NDVI as well as low (0.4304) and high (3.1431) EVI respectively and low ; (1.5692) and high (1.7895) SWIR. The values of Silhouette 0.5423, Davies–Bouldin 0.6799 and Calinski–Harabasz 215,201.56 indicate the presence of separable spectral regimes. If these don't show the classes originally proposed as Healthy, Moderately Stressed and Critically Stressed, it is because the rain data we have in this model are incomplete. Field calibration and re-sampling of spatial locations is still required.

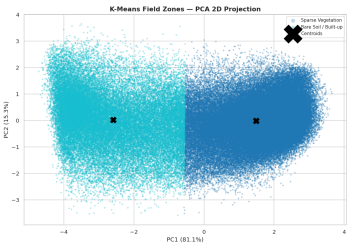
In component 4 the end-to-end pipeline is completely illustrated encompassing very high scores of the classifiers where the data is cleaned, encoded, trained, saved, loaded and simulated in the dashboard. The scores are indeed scores for tasks in which reconstruction of the composite label of the notebook is a genuine task. They don't support independently valid soil health as overlaps have been a component in building the target. Likewise, when the recurrent layer is fed one timestep, it should not be able to exhibit learnt temporal memory. This important interpretation is the magic key to getting seemingly good results translated into an easy sequence collection and external reference test plan.



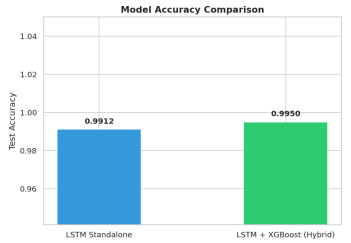
(a) Component 1: Anomaly-classifier performance summary.



(b) Component 2: Count and normalised test confusion matrices.



(c) Component 3: Two K-means field zones in the PCA projection.



(d) Component 4: LSTM and hybrid-model accuracy comparison.

Fig. 2. Representative notebook outputs from the four research components. Component 1 demonstrates high aggregate accuracy but zero anomaly-class precision, recall, and F1-score. Component 2 shows that test predictions were concentrated mainly in the veinal mottle virus and healthy classes. Component 3 identifies two statistically separable spectral regimes that require agronomic validation. Component 4 reports high derived-label classification accuracy, although genuine temporal forecasting remains to be validated.

V. Integrated Field Evaluation

A. Study Setting and Sampling

Evaluation will be carried out starting from the bench tests followed by marked-row and moving on to in chilli plantations. The actual needs discussed in the field (plantations) to Malagala, Padukka and Hanwella, Avissawella etc., guided by the Hector Kobbekaduwa Agrarian Research and Training Institute in these field visits cover points like crop-height variation, bagged and ground cultivation, row access, sensor insertion and camera adjustment. Stable identifiers will be given to the farms, visits, plants and spatial cells. Sampling will include

TABLE II
INTERPRETATION OF IMPLEMENTED NOTEBOOK EVIDENCE

C	Evidence	Defensible conclusion
1	Accuracy 0.96; ROC-AUC 0.8754; anomaly recall 0/9	Ranking signal exists; operating threshold and labels are inadequate
2	Validation near 0.98; mean test class accuracy 0.19	Strong domain/split failure; no field-readiness claim
3	$k = 2$; silhouette 0.5423; DB 0.6799	Two spectral regimes, not validated agronomic states
4	Very high derived-label classification; one timestep	Software pipeline works; independent soil prediction unproven

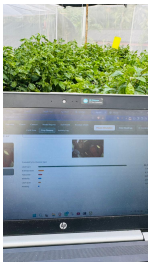
the various timing, illumination, crop development, visible conditions, moisture conditions and navigation obstruction. Repeat measurements will not be regarded as independent farms but instead will be stored as ”correlated” measurements.



(a) Chilli plants observed during the field visit.



(b) Rover-based examination of visible crop and soil conditions.



(c) AgroRover web dashboard demonstrated during field testing.

Fig. 3. Field observations and system demonstration conducted during the chilli-plantation visits. The activities supported the identification of practical requirements relating to crop-height variation, image acquisition, soil sensing, and AI-assisted crop-health monitoring.

B. Subsystem Metrics

Navigation testing utilizes surveyed reference points (or a higher grade receiver) and provides a report of position, heading and crosstrack error, completion time, stop frequency and percentage of samples that had degraded. Irrigation testing allows comparison of identical beds as it tracks flow, weather, initial bed conditions, emitter functioning, moisture stability, operator judgments and observations of stress, and production of the crop. Some measures for anomaly detection include confirmed (injected or observed) fault set, precision–recall AUC, class precision, recall, false-alert burden and detection delay [30] where precision and recall analysis is preferred under extreme imbalance.

In the case of vision evaluation, a set of field test images is held-out and the results of the rest of the images is reported including macro-F1, balanced accuracy, class-level recall, calibration error, selective accuracy, plant-level bootstrap intervals and open-set rejection. Visual criteria for diagnosis are recorded by reviewers, as is disagreement. The zone evaluation involves internal measurement and resampling stability, spatial and fragmentation, repeat-visit consistency, expert consensus and operability. Soil sensing is compared to multiple repeat and independent measurements of soil and forecasts are assessed sequentially against simple baselines. Usability sessions are used to determine if farmers are interpreting the confidence and warning, and comprehending the evidence in the zone and recommended actions.

C. End-to-End Experiments

Whenever possible, this integrated demo compares weather the existing manual practice, data from rover/marXsim coupled with the outputs from the different components, or the final flow of ev-idence is run. Field coverage, time until a suspect plant/zone was found, the percentage of observations which were of good position and quality, expert agreement of prioritisation, time between OBServation and REViewable RECommendation (ORRs), proportion of observations that did not return to a res non albus state (unresolved-alert rate), water/input use, and traceability of plant/zone actions completed. An integrated benefit should be greater than simply ”dashboards on a dashboard” or four dashboards in the same space.The integrated benefit should be more than just four dashboard screens in a single place.

VI. RELIABILITY, SAFETY, AND GOVERNANCE

The rover is a cyber-physical system, that means that only prediction performance is not enough. Multiple states of operation are covered by a layered state machine encompassing normal operation, degraded navigation, manual control, irrigation lockout and emergency stop. The motion commands are also watchdog timeouts; if a loss of control communication occurs then a stop is caused. Operating the pump needs a good mix of possible moisture, a valid state of the system, a limited duration and as much as possible a measured flow. Dry runs, blocked emitters, empty batteries, disconnections of the sensors, over and over GNSS jumps, camera failure and obsolete forecasts are deliberately tested.

VII. DISCUSSION

Not one of the main point is that four promising algorithms, alone, make an accurate autonomous agronomist. It’s the architecture, which makes their dependencies and their weaknesses apparent to you. Confidence in positions is now considered

as part of each observation, uncertainties carry over in fusion, uncertainties are separated from experts labels in unsupervised fusion, uncertainties are separated from forecasts in derived scores for soil. These boundaries are crucial because of the propagation of the errors. When coupled with easily placing a false prediction of disease as a treatment zone, and then treating without review, a false prediction of disease is more damaging when it is confidently mapped and managed as such.

VIII. Conclusion

The entire Smart Chilli Crop Intelligence System has been proposed in this paper which consists of Rover Path Planning and irrigation, artificial intelligence in cropping monitoring, field zone planning and soil decision support. The four pipelines will be complementing each other technically: trusted location will support the meaningful mapping of the images; image will support the plants with evidence; soil histories will provide the temporal and causal context, and zones provides a way to integrate the observations into coordinated work. Research conducted in the notebooks shows that implementation made good progress, but also shows several key restrictions that otherwise precludes the possibility of a statement to the extent of overall accuracy (or readiness for deployment). There are six steps to validation: calibration, subsystem testing, external field testing, validation with uncertainty-aware fusion, fault injection, and controlled workflow comparison that provide a defensible avenue for progress. It is envisaged that this should lead to a low cost, traceable and human supervised decision system that allows chilli farmers to allocate their attention and efforts, to where they are most needed.

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